Similarity

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected similarity-comparison record.

Trace points, change weights, preserve missing values and inspect duplicated features.

Keep one habit: Keep each comparison's target, candidates and rule version together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Confidence Is Not Correctness. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Confidence Is Not Correctness

Boundaries and 'Not Sure'

Learning from Examples

W1 · Trace Basic1's contributions

> Use the supplied records and return every highest score.

Your evidence record

- Basic1 reference: target red,round,smooth. A=red,round,rough; B=blue,square,smooth; C=red,square,rough. Feature order is colour,shape,texture. Give the declared points for an exact match and zero for a known mismatch; return every highest scorer. Pause final ranking and request the missing value when a required comparison field is unknown.

- For the red round smooth target and **candidate**s A, B and C in the lesson, record the colour, shape and texture contributions under one point per match. Give each total and identify every highest scoring **candidate**.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Recalculate after changing texture weight

> Keep candidate records fixed while comparing rule versions.

Your evidence record

- Keep the same target and **candidate**s but make texture worth 3 points, with colour and shape worth 1 each. Recalculate A, B and C. Explain what changed and what stayed the same.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Retain a missing texture value

> Keep known contributions without assigning an invented mismatch.

Your evidence record

- **Candidate** E is red and round but its texture is missing. A learner assigns zero for texture and declares E fully scored. Apply our classroom policy.

  State the known contributions, the missing information and what must happen before a final ranking.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Detect duplicated-feature weighting

> Explain what copying a colour column does to the effective rule.

Your evidence record

- A designer adds a second colour column that copies the first and gives both columns one point. They say no feature has been given extra importance because every column has equal **weight**.

  Explain the problem and propose a clearer rule.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Transfer the comparison to books

> Keep one first attempt before hints or feedback.

Read and explain independently

A target book has topic space, format comic and language English. **Candidate**s are J: space, prose, English; K: animals, comic, English; L: space, comic, Spanish. Give 1 point per exact match and return all tied highest scorers.

Then make topic worth 3 points while keeping the other **weight**s at 1. Report both sets of totals and winners. Explain why a winner may still fail a requirement to read English.

> **My independent explanation, trace and limits**
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Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Transfer the comparison to books

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Choose two useful item traits for **similarity**-based sorting and explain why each deserves its **weight**.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Explain a changed winner for **Similarity** and selected features. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Similarity** and selected features. Proposed explanation: “Similar has one objective meaning for every task”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A reader wants mystery and short books. A has both traits, B only mystery, C only short. Score one point per matching trait. The reader now says short is essential. Explain how the decision should change.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Book A has animals and adventure; B has animals only; C has neither. A reader wants both traits.

Rank by number of matching traits.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · A changed case

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Compare books using topic only. Book A is about space with a red cover; B is about space with a blue cover; C is about dogs with a red cover. Which matches A best?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 19 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A reader wants mystery and short books. A has both traits, B only mystery, C only short. Score one point per matching trait. The reader now says short is essential. Explain how the decision should change.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A target book has topic space, format comic and language English. **Candidate**s are J: space, prose, English; K: animals, comic, English; L: space, comic, Spanish. Give 1 point per exact match and return all tied highest scorers.

Then make topic worth 3 points while keeping the other **weight**s at 1. Report both sets of totals and winners. Explain why a winner may still fail a requirement to read English.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional craft-card matcher compares each **candidate** with a target.

Target: red, round, smooth. A: red,square,smooth. B: blue,round,smooth. C: red,round,rough.

Initially each exact **feature match** scores 1; return all highest scorers.

For the second comparison only, colour and shape matches score 1 each and a texture match scores 3.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/19

Similarity

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Target features

Compare colour, shape and texture separately.



2. Equal weights

A matches colour+texture=2; B shape+texture=2; C colour+shape=2.



3. Initial result

All three tie at 2.



4. Texture weight 3

New scores A=4, B=4, C=2; A and B remain tied.

CHANGE ONE THING

Keep original weights but change the target texture to rough.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Point to the matches supporting C's new score and explain why there is no tie.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Declare sorter **similarity** features, **weight**s and tie behavior before using scores to explain recommendations.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Declare sorter **similarity** features, **weight**s and tie behavior before using scores to explain recommendations.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional craft-card matcher compares each **candidate** with a target.
- Which labelled step matches this detail? New scores A=4, B=4, C=2; A and B remain tied.
- 1. Target features
- 2. Initial result
- 3. Texture **weight** 3
- 4. Equal **weight**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Texture **weight** 3 New scores A=4, B=4, C=2; A and B remain tied. **Similarity** is defined by the chosen representation and **weight**s.

The cards did not change when texture gained **weight**; the comparison rule did. Two high scores need not mean two identical cards.

A complete rule also says what to do with tied scores, rather than choosing whichever appears first by accident.

Changed-case question

- Changed-case practice: Keep original **weight**s but change the target texture to rough. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. C scores 3; A and B each score 1.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: C scores 3; A and B each score 1. C now matches all three target features; the others lose their smooth-texture match.

Repair and save your work

Review Confidence Is Not Correctness (ai-1/18). Goal: Distinguish a confidence claim from **evidence** that a **prediction** was correct. Short practice: Was the unknown label guaranteed correct because its confidence was **80%**?.

Return to this week and explain what you changed.

Review Boundaries and 'Not Sure' (ai-1/16). Goal: Identify a specific source of uncertainty and explain how it affects a **prediction**. Short practice: Which original block needs a better width before accepting or rejecting it?

What should the reviewer do?. Return to this week and explain what you changed.

Review Learning from Examples (ai-1/13). Goal: Fit a threshold by tracing **prediction**s and counting disagreements for every **candidate**. Short practice: Which new measured length would reveal disagreement between the **candidate**s?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Similarity: the case

AI ACADEMY / AI-1/19 · 01 · LOOK AT THE FACTS

Similarity: the case

The situation

A fictional craft-card matcher compares each candidate with a target.

Supplied evidence

Target: red, round, smooth. A: red,square,smooth. B: blue,round,smooth. C: red,round,rough.
Initially each exact feature match scores 1; return all highest scorers.
For the second comparison only, colour and shape matches score 1 each and a texture match scores 3.

By the end, I can calculate similarity under a declared feature rule and explain a changed ranking.

Follow the labelled picture with the matching explanation. Open the vector diagram

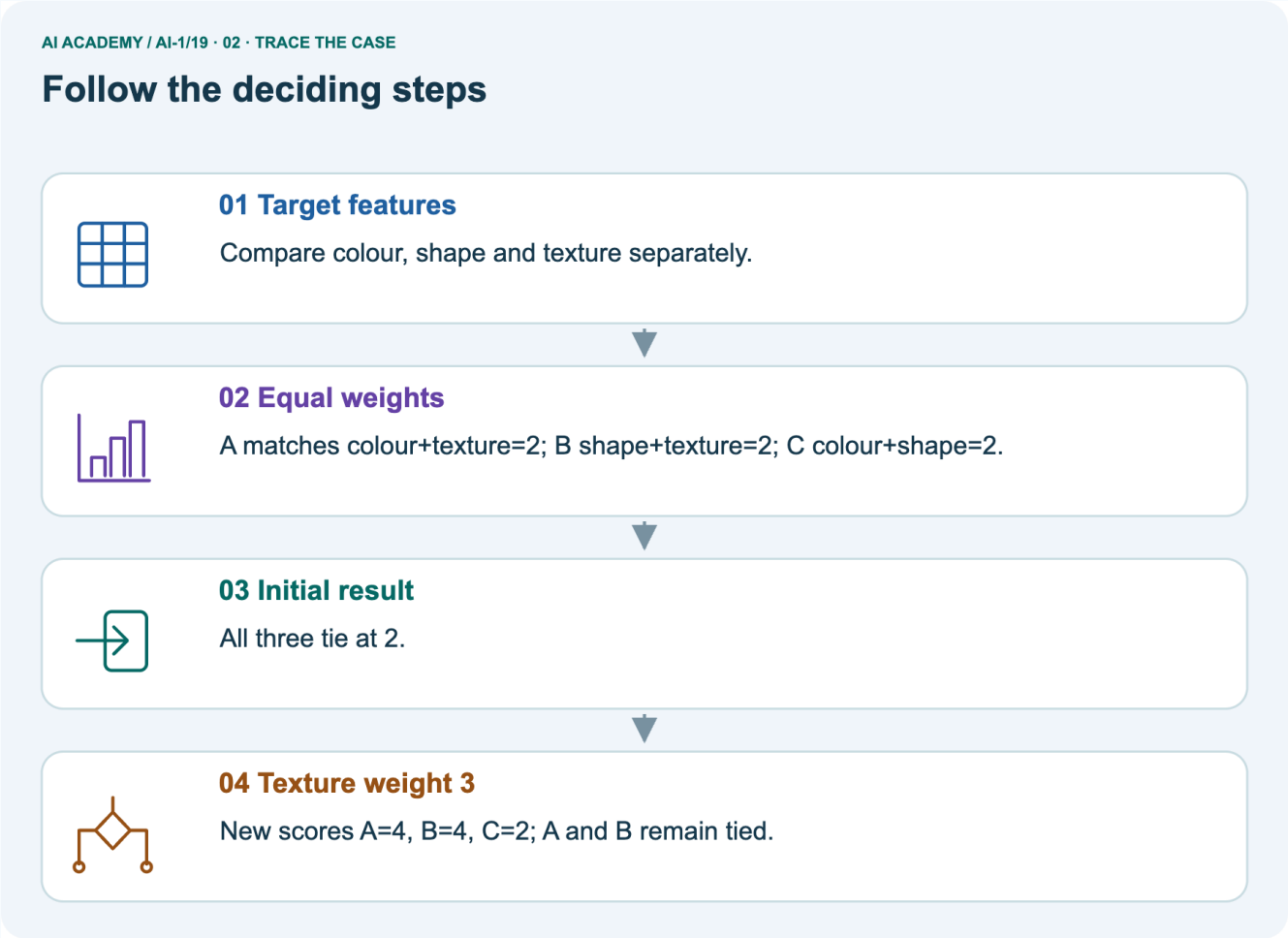
A fictional craft-card matcher compares each **candidate** with a target. Supplied facts: Target: red, round, smooth. A: red,square,smooth. B: blue,round,smooth. C: red,round,rough.

Initially each exact **feature match** scores 1; return all highest scorers. For the second comparison only, colour and shape matches score 1 each and a texture match scores 3.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Target features: Compare colour, shape and texture separately. → Equal **weight**s: A matches colour+texture=2; B shape+texture=2; C colour+shape=2. → Initial result: All three tie at 2. → Texture **weight** 3: New scores A=4, B=4, C=2; A and B remain tied.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Declared weights change similarity



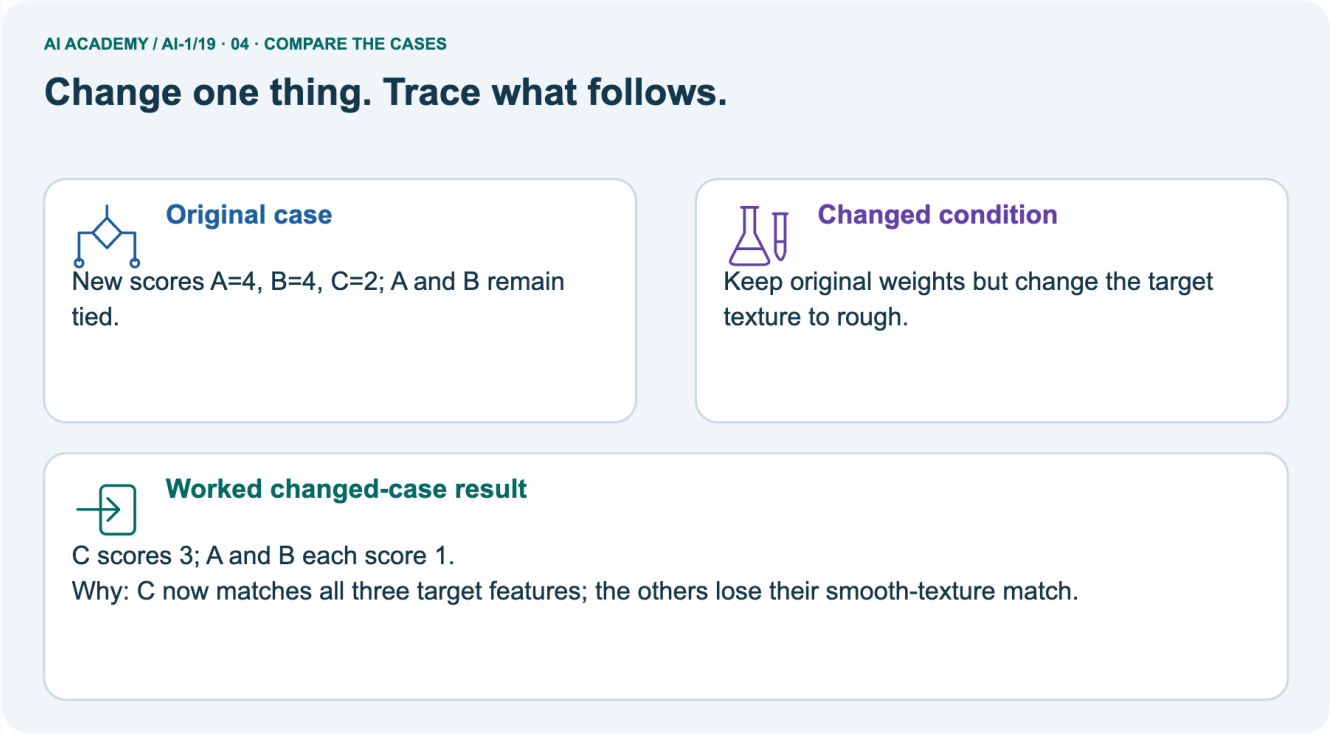
Follow the labelled picture with the matching explanation. Open the vector diagram

Declared **weight**s change **similarity**. A: All **weight**s 1 = 2, Texture **weight** 3 = 4. B: All **weight**s 1 = 2, Texture **weight** 3 = 4. C: All **weight**s 1 = 2, Texture **weight** 3 = 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

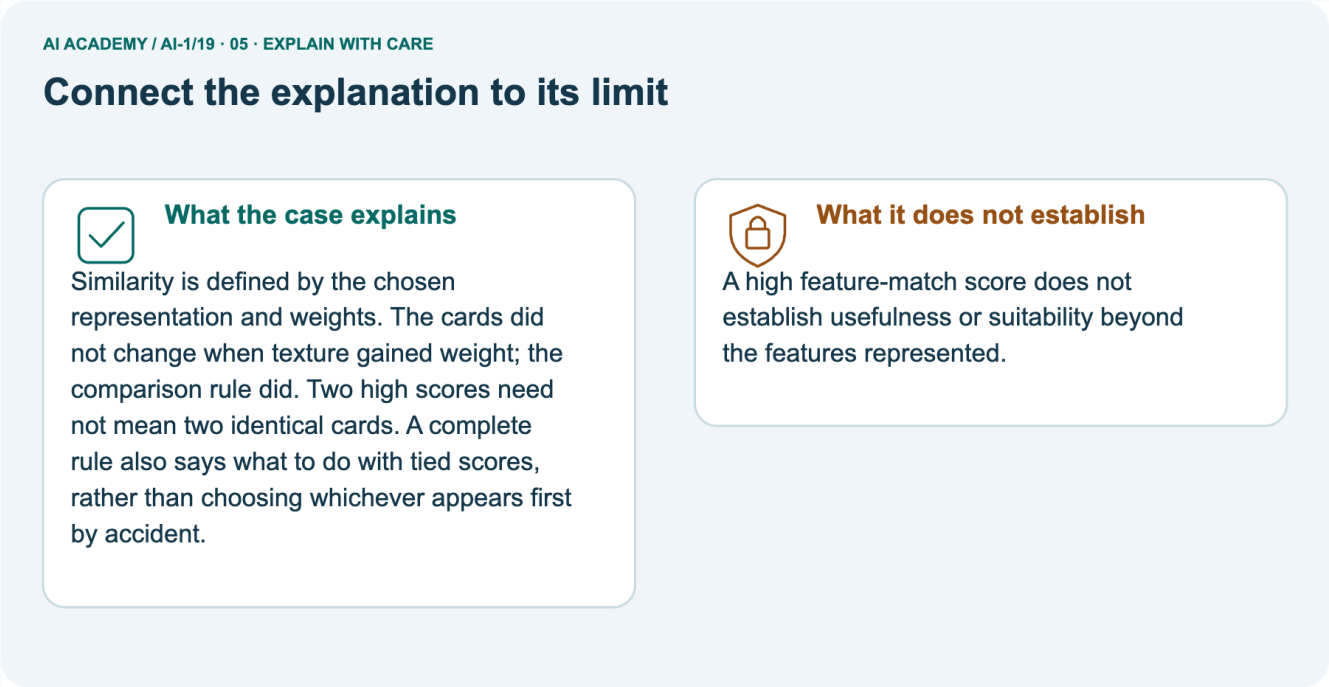
Original case: New scores A=4, B=4, C=2; A and B remain tied. Changed condition: Keep original **weight**s but change the target texture to rough. Expected result: C scores 3; A and B each score 1.

Reason: C now matches all three target features; the others lose their smooth-texture match.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

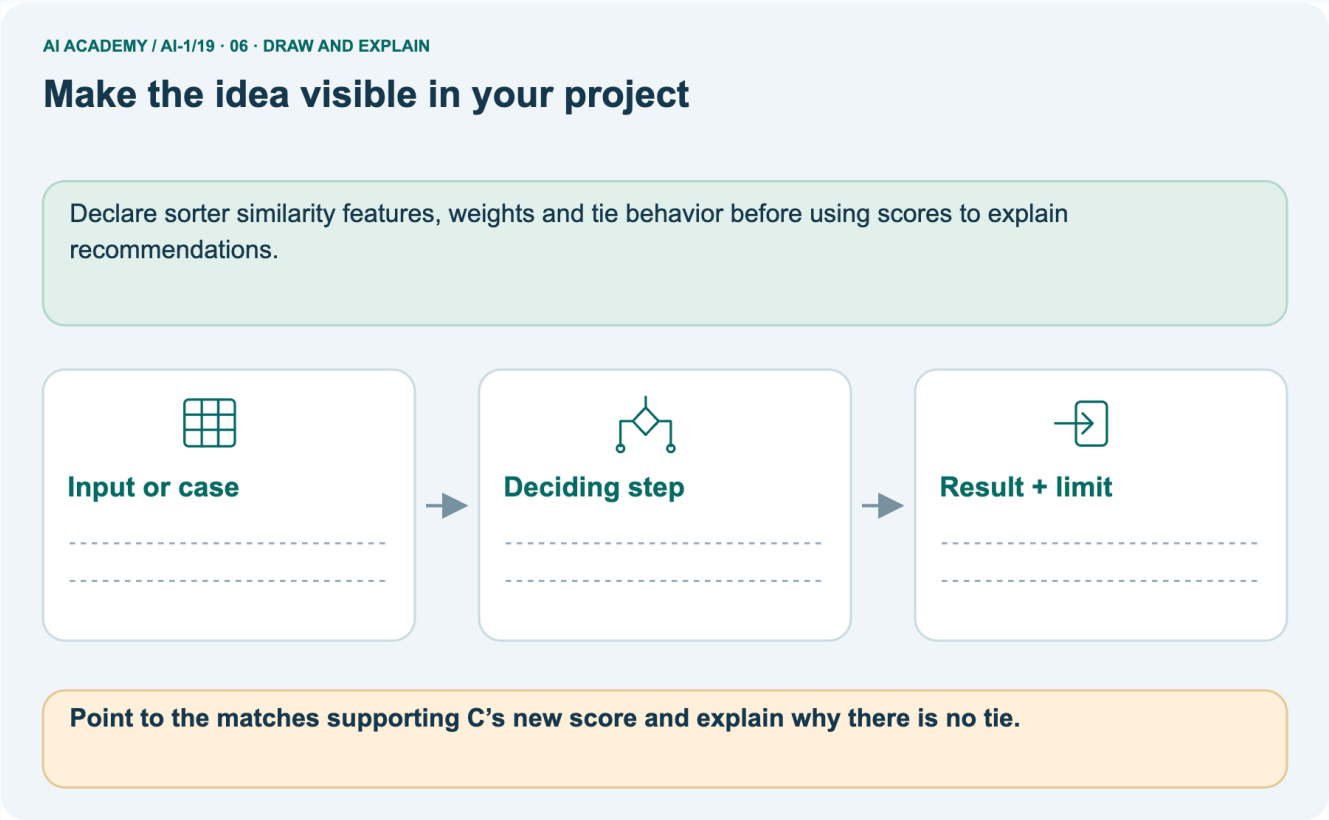
Similarity is defined by the chosen representation and **weight**s. The cards did not change when texture gained **weight**; the comparison rule did. Two high scores need not mean two identical cards.

A complete rule also says what to do with tied scores, rather than choosing whichever appears first by accident. Limit: A high feature-match score does not establish usefulness or suitability beyond the features represented.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Declare sorter **similarity** features, **weight**s and tie behavior before using scores to explain recommendations. Draw your **input**, deciding step, result and limit.

Supported question: Point to the matches supporting C's new score and explain why there is no tie.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Similarity: the case

02. Follow the deciding steps

03. Declared weights change similarity

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.